

Online Bin Covering Algorithm — Code Explanation

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Problem Description

In the online bin covering problem, items of sizes in $(0, 1]$ arrive one by one. When an item arrives, we must immediately decide which bin to place it into (or open a new bin). Each bin has capacity 1 and is *covered* when the total size inside reaches at least 1. The goal is to **maximize the number of covered bins**.

Our Approach

We implemented a greedy algorithm that processes items in the order they arrive. For each item, we try the following strategies in priority order:

- **Immediate coverage:** First, we check if placing the item into any existing bin would cover it (sum becomes ≥ 1). If multiple bins qualify, we choose the one that creates the least waste (closest to exactly 1). This is our top priority because covering a bin immediately is always beneficial.
- **Large items:** If the item size is greater than 0.5 and we cannot cover a bin, we place it into the smallest open bin. Large items cannot pair with another large item (their combined sum would exceed 1), so they must eventually pair with small items. Putting them in the smallest bin keeps bins balanced and leaves room for future small items to complete the coverage.
- **Best-fit for small items:** For small items (≤ 0.5), we use best-fit: we place the item into the bin with the largest current sum that still has room. This maximizes the chance of pushing bins closer to coverage.
- **New bin:** If none of the above apply, we open a new bin with this item.

We use a small constant $\text{EPS} = 1\text{e-}9$ when comparing sums to 1 to account for floating-point rounding errors.

Data Structures

We maintain two lists:

- **open_bins:** each entry stores the bin's current sum and a list of item indices inside it.
- **covered:** stores the item-index lists of bins that have been covered (moved here once their sum reaches ≥ 1).

Complexity

The algorithm runs in $O(n \cdot m)$ time where n is the number of items and m is the number of open bins at any given point. In the worst case $m = O(n)$, giving $O(n^2)$ overall, but in practice m remains small. Space complexity is $O(n)$.